ELECTRICAL ENGINEERING
UNIT 1: EVOLUTION OF ELECTRICAL POWER & ENGINEERING EDUCATION
Curriculum: Lovely Professional University | Department of Electrical Engineering
Reference: Electric Power Systems by C. L. Wadhwa / IEEE Power History Archives

>> 1.1 Historical Genesis: From Static Electricity to Bulk Power Networks

The discipline of Electrical Engineering traces its lineage through revolutionary discoveries:
- 1800 (Alessandro Volta): Invention of the voltaic pile, providing the first continuous chemical source of direct current (DC).
- 1831 (Michael Faraday): Discovery of Electromagnetic Induction (Faraday's Law: e = -dPhi/dt), enabling mechanical generation of electricity.
- 1882 (Thomas Edison): Commissioning of the Pearl Street Station in New York City, the world's first commercial central DC power plant delivering 110V DC to 59 customers.
- The Fundamental Limitation of Low-Voltage DC: Edison's low-voltage DC system suffered severe I^2*R transmission losses (P_loss = I^2*R). Transmitting power over just 1.5 km required enormous copper conductors.

>> 1.2 The "War of the Currents": Edison (DC) vs. Westinghouse & Tesla (AC)

The struggle between Direct Current (DC) and Alternating Current (AC) defined modern power engineering:
- Nikola Tesla patented the polyphase AC system, polyphase induction motor, and multi-phase transformers in 1887-1888.
- George Westinghouse commercialized Tesla's AC patents using William Stanley's practical step-up/step-down transformer design.
- The Mathematical Supremacy of AC: Transmitted power P = V * I. To deliver power P at voltage V, current is I = P / V.
- Transmission loss is P_loss = I^2 * R = (P / V)^2 * R. Stepping up voltage by a factor of 10 decreases line current by 10x and reduces copper losses by 100x!
- 1895: Niagara Falls AC hydroelectric plant proved that polyphase AC could transmit bulk electrical power over 40 km to Buffalo, cementing AC dominance globally.

>> 1.3 HVAC vs. HVDC Transmission Systems

While AC won the historic battle, modern bulk transmission leverages both HVAC and High-Voltage DC (HVDC):
- High-Voltage AC (HVAC): Global standard for regional grids (132 kV, 220 kV, 400 kV, 765 kV). Utilizes inexpensive step-up and step-down transformers.
- High-Voltage DC (HVDC): Essential for long-distance bulk transmission (> 600 km overhead, > 50 km submarine/underground) and asynchronous grid interconnection.
- Converter Technologies: Line Commutated Converters (LCC using high-power thyristors)

and Voltage Source Converters (VSC using IGBTs).
- HVDC eliminates charging current losses of long AC cables, overcomes stability distance limits, and prevents cascading power grid blackouts.

>> 1.4 Evolution of Electrical Engineering Curricula

The academic paradigm has transformed across four major eras:
- Era 1 (1890s-1940s): Heavy power engineering, DC/AC rotating machinery, synchronous alternators, overhead transmission towers.
- Era 2 (1950s-1980s): Solid-state physics, power electronics (SCRs, triacs), analog control theory, power system stability and protection relays.
- Era 3 (1990s-2010s): Digital Signal Processors (DSP), Microcontroller-based embedded drives, SCADA, static VAR compensators (FACTS).
- Era 4 (2020s-Present): Renewable Microgrids, Smart Grids, Electric Mobility (EV powertrains), Battery Management Systems (BMS), AI in Power Systems.

>> 1.5 Global & Indian Energy Landscape

India boasts the world's third-largest synchronous national electricity grid (One Nation, One Grid, One Frequency):
- Central Electricity Authority (CEA) and Ministry of Power govern national grid planning and regulatory frameworks.
- Installed generation capacity exceeds 430 GW, transitioning rapidly from thermal dominance toward renewable generation (500 GW non-fossil target by 2030).
- National Smart Grid Mission (NSGM): Deployment of Advanced Metering Infrastructure (AMI), automated fault location, and rooftop solar integration.

DETAILED WORKED EXAMPLES & ENGINEERING COMPUTATIONS

>> Example 1.1: AC vs. DC Transmission Loss and Conductor Sizing

Statement: A power generation station delivers 500 MW of active power over a distance of 400 km. The transmission line has a loop resistance of R = 8 Ohms. Assume a power factor of unity (cos Phi = 1.0).
Calculate the line current, line copper loss (I^2*R), and the percentage transmission loss when transmitting power at:
(a) Medium Voltage: V_1 = 66 kV
(b) Extra High Voltage: V_2 = 400 kV
(c) Ultra High Voltage: V_3 = 765 kV

Step-by-Step Solution:

1. Basic Transmission Equations:
- Transmitted active power: P = sqrt(3) * V_L * I_L * cos(Phi)
- Line current: I_L = P / (sqrt(3) * V_L * cos(Phi))
- Total three-phase line power loss: P_loss = 3 * I_L^2 * R_ph
- If R is the total loop resistance across the conductors, P_loss = I_L^2 * R.

2. Case (a): Transmission at V_1 = 66 kV (66,000 V):
- Line current: I_1 = (500 * 10^6) / (sqrt(3) * 66,000 * 1.0) = 4,373.86 A
- Total copper loss: P_loss1 = (4,373.86)^2 * 8 = 153,046,000 W = 153.05 MW
- Percentage loss = (P_loss1 / P) * 100 = (153.05 / 500) * 100 = 30.61%
- Observation: Over 30% of power is lost as heat; conductor would melt under thermal stress!

3. Case (b): Transmission at V_2 = 400 kV (400,000 V):
- Line current: I_2 = (500 * 10^6) / (sqrt(3) * 400,000 * 1.0) = 721.69 A
- Total copper loss: P_loss2 = (721.69)^2 * 8 = 4,166,700 W = 4.17 MW
- Percentage loss = (P_loss2 / P) * 100 = (4.17 / 500) * 100 = 0.834%
- Observation: Stepping up to 400 kV slashes line losses from 30.6% down to just 0.83%!

4. Case (c): Transmission at V_3 = 765 kV (765,000 V):
- Line current: I_3 = (500 * 10^6) / (sqrt(3) * 765,000 * 1.0) = 377.35 A
- Total copper loss: P_loss3 = (377.35)^2 * 8 = 1,139,150 W = 1.14 MW
- Percentage loss = (1.14 / 500) * 100 = 0.228%
- Loss reduction factor from 66 kV to 765 kV: (765 / 66)^2 = (11.59)^2 = 134.35 times lower losses!

>> Example 1.2: HVDC vs. HVAC Break-Even Economic & Technical Analysis

Statement: A power planning commission evaluates transmitting 2,000 MW from an offshore wind corridor over a distance of L km to a major industrial hub. The cost data is as follows:
- HVAC Line Capital Cost: C_line_ac = Rs. 1.8 Crore per km
- HVAC Terminal Substations (both ends): C_term_ac = Rs. 400 Crore total
- HVDC Bipolar Line Capital Cost: C_line_dc = Rs. 1.1 Crore per km (requires only 2 poles vs 3 phases)
- HVDC Converter Stations (both ends): C_term_dc = Rs. 1,600 Crore total
(a) Calculate the break-even transmission distance (L_break) in kilometers.
(b) If the transmission route is 1,200 km, determine the total capital savings achieved by selecting HVDC over HVAC.

Step-by-Step Solution:
1. Formulation of Total Capital Cost Equations:
- Total HVAC Cost: Cost_AC(L) = C_term_ac + (C_line_ac * L) = 400 + (1.8 * L) [in Rs. Crore]
- Total HVDC Cost: Cost_DC(L) = C_term_dc + (C_line_dc * L) = 1,600 + (1.1 * L) [in Rs.

Crore]

2. Setting Cost_AC(L_break) = Cost_DC(L_break):
- 400 + 1.8 * L_break = 1,600 + 1.1 * L_break
- (1.8 - 1.1) * L_break = 1,600 - 400
- 0.7 * L_break = 1,200
- L_break = 1,200 / 0.7 = 1,714.29 km (for purely overhead line).

3. Subsea / Underground Cable Correction:
- If 100 km of the link is subsea cable where AC charging current necessitates reactive compensation every 20 km:
- AC subsea line cost is Rs. 9.5 Crore/km, while DC subsea cable is Rs. 4.2 Crore/km.
- Under subsea cable conditions, the break-even distance drops dramatically to between 50 km and 80 km.

4. Capital Expenditure Analysis for 1,200 km Overhead Route with Right-of-Way (RoW) Factor:
- When factoring RoW land acquisition (HVDC corridor is 40m wide vs 75m for HVAC):
- Adjusted HVAC cost = Rs. 2.1 Crore/km; Adjusted HVDC cost = Rs. 1.25 Crore/km.
- Revised L_break = (1,600 - 400) / (2.1 - 1.25) = 1,200 / 0.85 = 1,411.76 km.

>> Example 1.3: Transformer Efficiency & Annual Carbon Offset Calculation

Statement: A 100 MVA, 400/132 kV, 3-phase grid step-down transformer has an iron core loss (no-load loss) of P_i = 45 kW and full-load copper loss of P_cu = 180 kW.
(a) Determine the transformer efficiency at full load operating at 0.85 power factor lagging.
(b) Determine the load kVA at which maximum efficiency occurs, and calculate the maximum efficiency.
(c) An older transformer had P_i = 90 kW and P_cu = 320 kW. Calculate the annual energy saved (kWh) by the modern transformer operating 24 hours/day, 365 days/year at an average load factor of 75%.

Step-by-Step Solution:
1. Full Load Efficiency at cos(Phi) = 0.85:
- Active Output Power: P_out = S * cos(Phi) = 100,000 kVA * 0.85 = 85,000 kW
- Total Losses: P_loss = P_i + P_cu = 45 kW + 180 kW = 225 kW
- Input Power: P_in = P_out + P_loss = 85,000 + 225 = 85,225 kW
- Efficiency eta = (P_out / P_in) * 100 = (85,000 / 85,225) * 100 = 99.736%

2. Load for Maximum Efficiency (x_max):
- Maximum efficiency occurs when variable copper loss equals constant iron loss: x^2 * P_cu = P_i
- Fraction of full load: x_max = sqrt(P_i / P_cu) = sqrt(45 / 180) = sqrt(0.25) = 0.50 (i.e., 50% load)

- Load at max efficiency: S_max_eta = 0.50 * 100 MVA = 50 MVA.
- Output at 50% load: P_out_50 = 50,000 * 0.85 = 42,500 kW
- Total losses at 50% load: P_loss_50 = P_i + (0.50)^2 * P_cu = 45 + 0.25 * 180 = 45 + 45 = 90 kW
- Max Efficiency: eta_max = (42,500 / (42,500 + 90)) * 100 = (42,500 / 42,590) * 100 = 99.789%

3. Annual Energy Savings Comparison:
- Operating hours per year: T = 24 * 365 = 8,760 hours.
- Average load fraction: x = 0.75 (75% load).
- Loss of Modern Transformer: P_loss_mod = 45 + (0.75)^2 * 180 = 45 + (0.5625 * 180) = 45 + 101.25 = 146.25 kW
- Loss of Older Transformer: P_loss_old = 90 + (0.75)^2 * 320 = 90 + (0.5625 * 320) = 90 + 180 = 270.00 kW
- Power saved: Delta_P = 270.00 - 146.25 = 123.75 kW
- Annual Energy Saved = 123.75 kW * 8,760 h = 1,084,050 kWh (over 1.08 Million Units/year)!
- At an average grid tariff of Rs. 7.50/kWh, annual monetary savings = Rs. 81.3 Lakhs/year.

>> Example 1.4: Grid Frequency Nadir and RoCoF Dynamics During Generator Outage

Statement: A regional power grid has an aggregated operating capacity of S_n = 40 GW (40,000 MVA) operating at 50.0 Hz. Suddenly, a major thermal power station generating Delta_P = 1,200 MW trips offline.
(a) Calculate the initial Rate of Change of Frequency (RoCoF = df/dt) when the grid inertia constant is H = 4.5 seconds (conventional thermal/hydro grid).
(b) Calculate the RoCoF if high penetration of inverter-based solar/wind resources reduces effective grid inertia to H_eff = 2.0 seconds.
(c) Determine the critical response time for Fast Frequency Response (FFR) battery energy storage systems (BESS) to prevent frequency from dropping below the under-frequency load shedding (UFLS) trigger of 49.2 Hz.

Step-by-Step Solution:
1. Power System Swing Equation for Aggregated System Frequency:
- The fundamental relationship between power imbalance and frequency derivative is:
(2 * H / f_0) * (df/dt) = - Delta_P / S_n
- RoCoF: df/dt = - (Delta_P * f_0) / (2 * H * S_n)

2. Case (a): Traditional High-Inertia Grid (H = 4.5 s):
- df/dt = - (1,200 * 50.0) / (2 * 4.5 * 40,000) = - (60,000) / (360,000) = - 0.1667 Hz/s
- Interpretation: Frequency drops at a manageable rate of 0.167 Hz per second, giving thermal governors (3-5 second response) ample time to ramp up.

3. Case (b): Low-Inertia Renewable Grid (H_eff = 2.0 s):

- df/dt = - (1,200 * 50.0) / (2 * 2.0 * 40,000) = - (60,000) / (160,000) = - 0.3750 Hz/s
- Interpretation: The rate of decay is 2.25 times faster. Without synthetic inertia, frequency drops dangerously fast.

4. Critical Time to Reach UFLS Threshold (49.2 Hz):
- Permissible frequency deviation: Delta_f = 50.0 - 49.2 = 0.8 Hz
- In Case (a): t_crit = Delta_f / |df/dt| = 0.8 / 0.1667 = 4.80 seconds.
- In Case (b): t_crit = Delta_f / |df/dt| = 0.8 / 0.3750 = 2.13 seconds.
- Engineering Action: Grid code mandates BESS battery injection within 500 ms to arrest RoCoF and preserve grid stability.